

Name: _____

Solving Equations Word Problems

Directions: Read each word problem carefully. Write an equation to model the situation and solve for x . Show your work

THE ROBOT TASK MANAGER

- 1.) **Package Delivery Task:** The robot delivers 5 packages every trip. After completing x trips, it has delivered 35 packages. How many trips did the robot make?

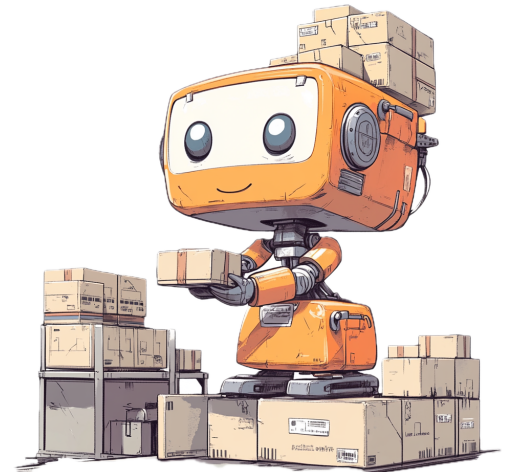
- 2.) **Weight Sorting Task:** A robot sorts objects weighing 2 kilograms each. It sorts x objects, and the total weight is 14 kilograms. How many objects did the robot sort?

- 3.) **Cleaning Task:** The robot's cleaning speed is 3 square meters per minute. It has cleaned 18 square meters so far. How many minutes has the robot spent cleaning?

- 4.) **Oil Refill Task:** A robot starts its shift with 8 liters of oil. It consumes oil at a rate of 2 liters per hour. After working for x hours, it has 4 liters left. How many hours has the robot worked?

- 5.) **Battery Recharging Task:** A robot starts with a battery charge of 50%. It uses 10% of its battery every hour. After x hours, the battery is down to 20%. How long has the robot been working?

- 6.) **Assembly Line Task:** A robot assembles 4 products every hour. After assembling for x hours, it has completed 28 products. How many hours did the robot work?



ANSWER KEY

1. Package Delivery Task:

Equation: $5x = 35$

Solution: $x = 7$

2. Weight Sorting Task:

Equation: $2x = 14$

Solution: $x = 7$

3. Cleaning Task:

Equation: $3x = 18$

Solution: $x = 6$

4. Oil Refill Task:

Equation: $8 - 2x = 4$

Solution: $x = 2$

5. Battery Recharging Task:

Equation: $50 - 10x = 20$

Solution: $x = 3$

6. Assembly Line Task:

Equation: $4x = 28$

Solution: $x = 7$